

Le Nhu Dat

lenhudat181104@gmail.com | 0343683245 | [linkedin.com/in/lenhudat-ai-engineer](https://www.linkedin.com/in/lenhudat-ai-engineer) | github.com/lndat18

SUMMARY

AI Engineer focused on LLM agents and RAG systems that automate knowledge-heavy work. 1 year building RAG and agent pipelines end to end: hybrid retrieval, reranking, answer verification, evaluation and observability.

PROJECTS

Vietnamese Legal QA – Production RAG

github.com/lndat18/production-legal-qa-rag

- Built an end-to-end RAG chatbot over 6 Vietnamese labor, insurance and tax laws (2,228 structure-aware chunks) that answers with Article/Clause citations, replacing manual legal text lookup
- Designed a hybrid retrieval pipeline: dense + BM25 with Reciprocal Rank Fusion, HyDE, citation-aware candidate search, optional MMR, and a local Vietnamese GPU reranker
- Added a fail-closed verification layer (deterministic citation and number checks + an independent LLM evidence judge, at most one repair): on the 157-question testset it refused 9% of cases lacking supported evidence and repaired 12% before release
- Engineered serving reliability: Redis answer/retrieval cache with single-flight, admission control, per-user rate limits, conversation follow-up handling, and throttling across multiple Groq API keys
- Shipped an OpenAI-compatible REST API (FastAPI) with an OpenWebUI chat interface, deployed on a single host with Docker Compose and Cloudflare Tunnel, automated backups, Langfuse traces and Prometheus/Grafana metrics, and a GitHub Actions CI running 1,700+ tests with Ruff and mypy
- Built a RAGAS evaluation pipeline: generated 203 synthetic cases, reviewed them down to a 157-question golden testset (142 single-hop, 15 multi-hop), and scored Context Precision 0.90, Context Recall 0.87, Faithfulness 0.83 and Answer Relevancy 0.43, with checkpointed, resumable runs that handle API quota limits

LLM & Retrieval: LangChain, Groq (gpt-oss-120b, gpt-oss-20b), Hugging Face, Pinecone, BM25, PyTorch

Backend & Deployment: Python, FastAPI, Redis, PostgreSQL, OpenWebUI, Docker, Cloudflare Tunnel

Observability: Langfuse, Prometheus, Grafana

Evaluation & CI/CD: RAGAS, pytest, Ruff, mypy, GitHub Actions

Agentic GraphRAG for Legal Q&A — In Progress

github.com/lndat18/production-legal-agentic-graph-rag

- Building a LangGraph agent on top of the RAG system above that plans, retrieves, calls tools and verifies, to answer legal questions spanning multiple clauses and articles
- Built a legal knowledge graph in Neo4j that models cross-references between articles, clauses and points, so retrieval can follow legal citations across documents
- Designing three cooperating retrieval agents (retrieval planner, hybrid retriever, graph retriever) that decompose a request into subqueries and gather connected provisions
- Designing a personal income tax agent: a deterministic calculator owns the arithmetic while the LLM extracts parameters, with per-bracket breakdowns and legal citations, DOCX/PDF form drafts, effective-date checks, and user confirmation before any external action
- Integrating MCP to expose tools and data sources to the agent and Jev for routing decisions, evaluated with DeepEval and RAGAS on tool-call accuracy, task success rate, numeric error and cost

Agent Orchestration & Graph: LangGraph, Neo4j

LLM & Retrieval: LangChain, Groq (gpt-oss-120b, gpt-oss-20b), Hugging Face, Pinecone, BM25, PyTorch

Backend & Deployment: Python, FastAPI, Redis, PostgreSQL, OpenWebUI, Docker, Cloudflare Tunnel

Observability: Langfuse, Prometheus, Grafana

Evaluation & CI/CD: RAGAS, pytest, Ruff, mypy, GitHub Actions, DeepEval

Tools & Routing: MCP, Jev

EXPERIENCE

AI Engineer Intern

Ho Chi Minh City, Vietnam

TMA Solutions – leading Vietnamese software company (4,000+ engineers, 30+ countries) March 2026 – May 2026

- Developed a Sign Language Recognition solution for the hearing-impaired using CorrNet and MediaPipe keypoints
- Improved the preprocessing pipeline to handle variable-length sequences and missing keypoints, raising accuracy from 90% to 94% and reducing word error rate (WER) from 9% to 7% on the public How2Sign dataset
- Applied the approach validated on How2Sign to the company's own sign language dataset for the same recognition task, retraining the model from scratch and tuning its architecture and hyperparameters for that data

EDUCATION

TON DUC THANG UNIVERSITY

Ho Chi Minh City, Vietnam

Bachelor of Computer Science

2022 – 2026

GPA: 3.06/4.00 Classification: Good

CERTIFICATIONS

- [IBM RAG and Agentic AI](#) — IBM
- [IBM Generative AI Engineering Professional Certificate](#) — IBM
- [LLMOps \(Large Language Model Operations\)](#) — Duke University
- [MLOps \(Machine Learning Operations\)](#) — Duke University
- [AIO2025](#)
- [Retrieval Augmented Generation \(RAG\)](#) — DeepLearning.AI
- [Natural Language Processing Specialization](#) — DeepLearning.AI
- [Deep Learning Specialization](#) — DeepLearning.AI
- [Machine Learning Specialization](#) — DeepLearning.AI
- [Introduction to Git and GitHub](#) — Google

ADDITIONAL INFORMATION

Programming Languages: Python, C, Java

Language Skills: [IELTS 5.5](#) (CEFR B2)